SOMAIYA
VIDYAVIHAR

K J Somaiya Institute of Technology
An Autonomous Institute permanently affiliated to University of Mumbai.

Introduction of carpentry hand tools

tools

Carpentry hand tools are listed as per the following:

1 Marking and testing tools

(e.g) scriber, spirit leve.l, try squares, marking gauge etc.,

2 Holding and supporting tools

(e.g) mitre box, work table, clamps etc.,

3 IV]easuring tools

(e.g) Foot rule, tape rule, caliper etc.,

4 Cutting tools

(e.g) saws, chise.ls etc..,

5 Planning tools

(e.g) Planes, spoke shave etc.

6 Boring tools

(e.g) Hand drill, auger, bits, twist bit, etc.,

7 Striking tools

(e.g), Hammers, mallet etc.,

8 Driving tools

(e.g), Screw driver, spanner s etc.,

9 Miscellaneous tools

(e.g) Punches, Pincer etc.,

10 Abrasion tools

(e.g), Files, oil stone etc.,

Classification and uses of marking tools

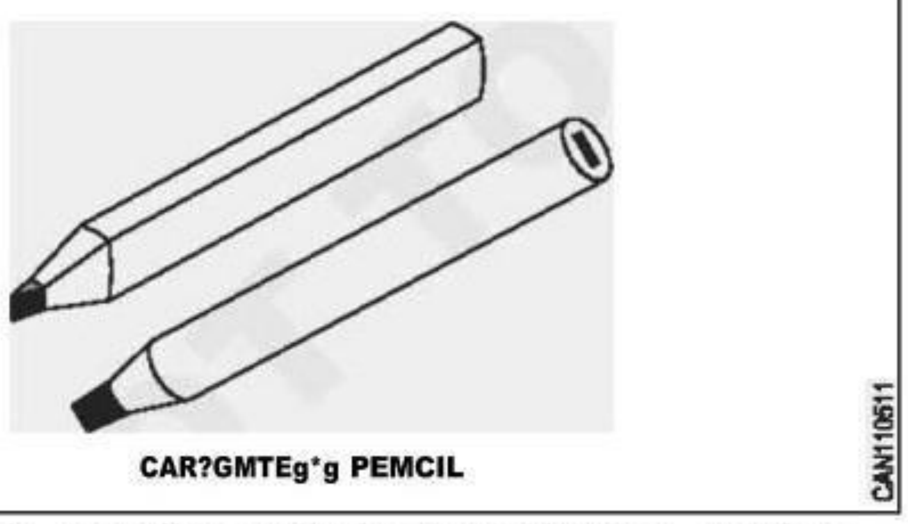
Objectives: At the end of this lesson you shall be able to
- state **the diPerent** type **of marking tools**
- explain the use of marking tools
- **brief constructional** features of marking tools.

Marking off or layout is carried out to indicate the location of operation to be done and provide guidance during sequence of operations.

– IV] arking out is done with pencil or.scriber etc..,

Carpenter's pencil (Fig 1)

Fig 1



Carpenter's pencil usually is an oval cross-section

– It is sharpened with a chisel

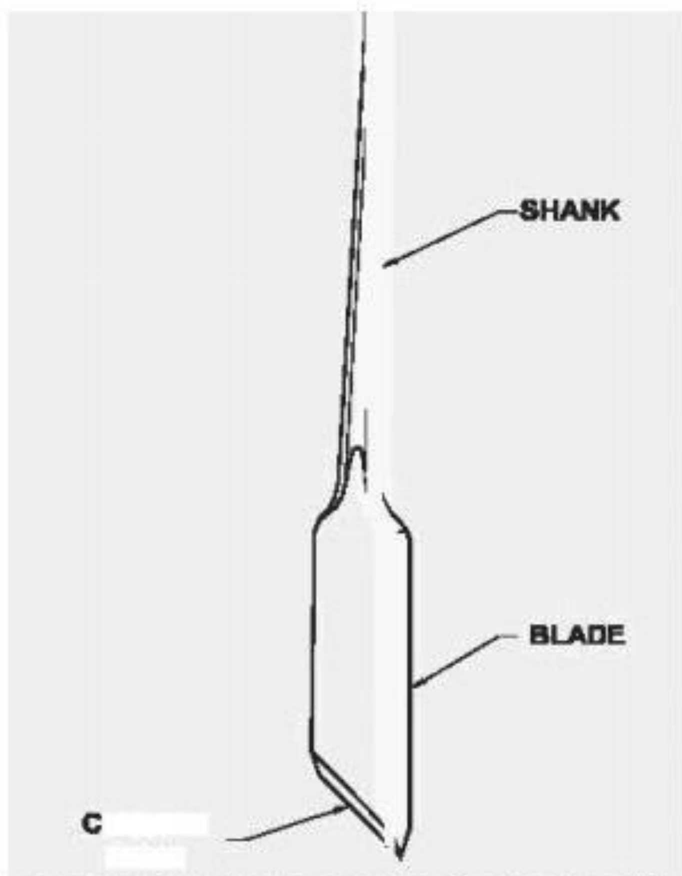
– The pencil is riot used for an accurate wo.rk.

– Suitable pencil hardness for marking out o'n 'HB', 'H' and'F'

Marking knife (Fig 2)

It is made of.steel fashioned to a point at one end and a sharp blade at the other end to form a cutting edge. The blade or knife is used for marking cut lines where a vertical s.houlder is to be cut with a saw or chisel. The point is used for marking distances and scribing lines.

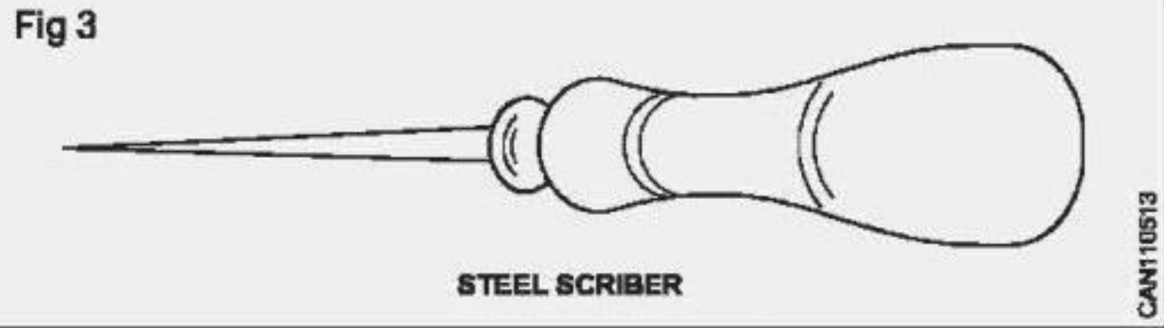
Fig 2



MARKING KNIFE (OR SCRIBER KNIFE)

Steel scriber (Fig 3)

Fig 3

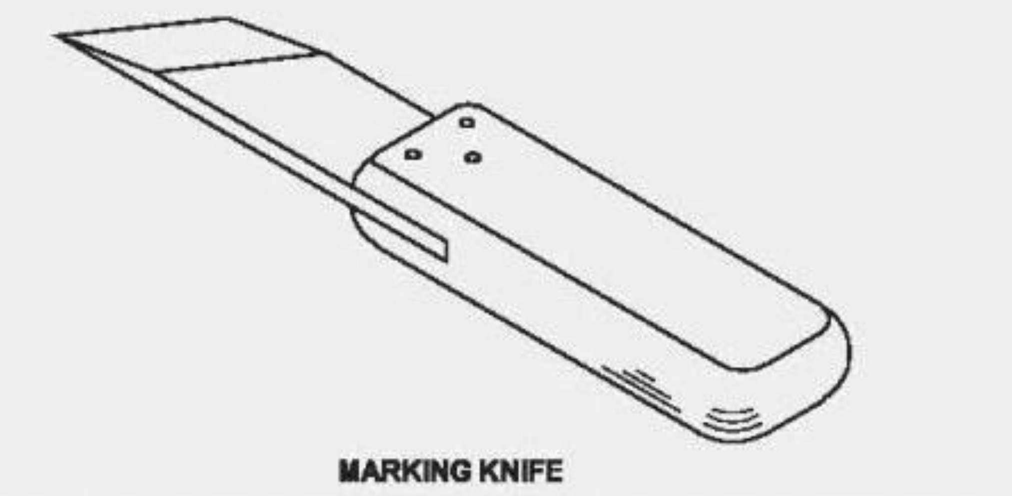


A steel scriber should be sharp at its point. It is used for scribing lines on which a chisel cut or a saw cut is made

The scriber should not be used as an awl. Do not strike the handle with a hammer.

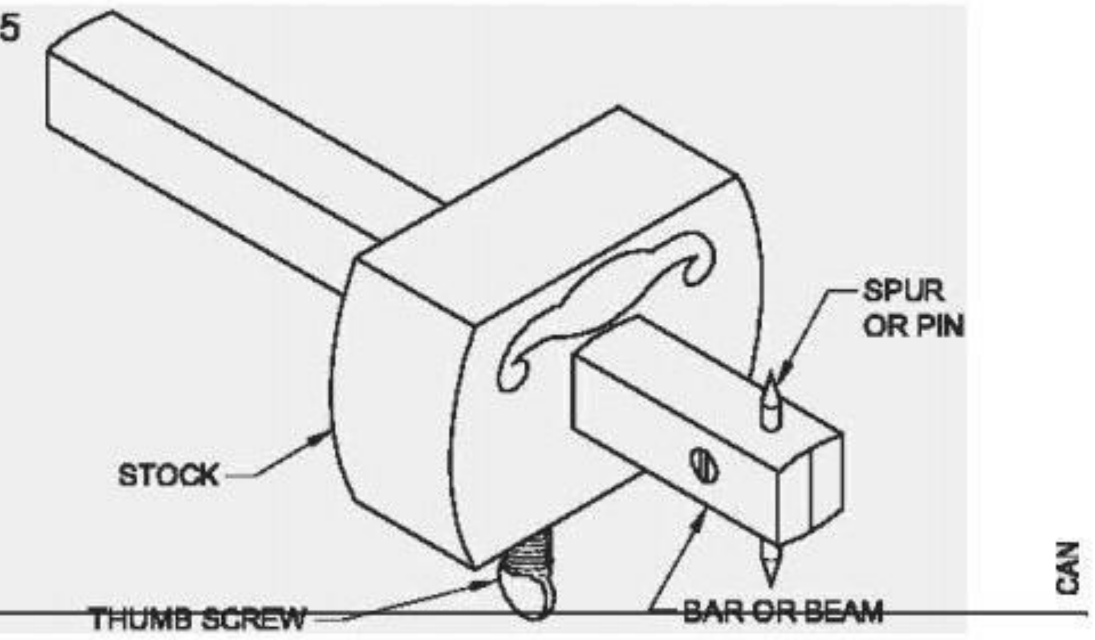
Marking knife is also used for marking and scribing . It is a steel blade fixed in a wooden handle. it serves the same purpose as that of scriber. (Fig 4)

Fig 4



Marking gauge (Fig 5)

Fig 5



Making gauge is used for marking lines parallel to a face and edge (e.g) gauging width and thickness.

The marking gauge can be made of wood or steel. The gauge consists of square, wooden bar or beam on which wooden block or stock is sliding. This block can be fastened at any required measurement by use ofa thumb screw.

In better form of gauges the stock is protected from wear by a piece of brass set flush with surface. The bar is graduated in millimeter and provided with a spur or steel point at the end. It is always advisable to measure the distance from the spur to the face of the block, with an ordinary rule

The gauge is set by loosing the screw and the stock is shifted to the distance required from the spur. Measurement are taken from a rule.

After setting the screw is tightened while gauging the stock is firmly help against the wood and pushed in forward direction. (Fig 6)

Mortise gauge (Fig 7)

A mortise gauge is a making gauge with two spurs. The two spurs can be spaced at different distances and mark two parallel lines at a time

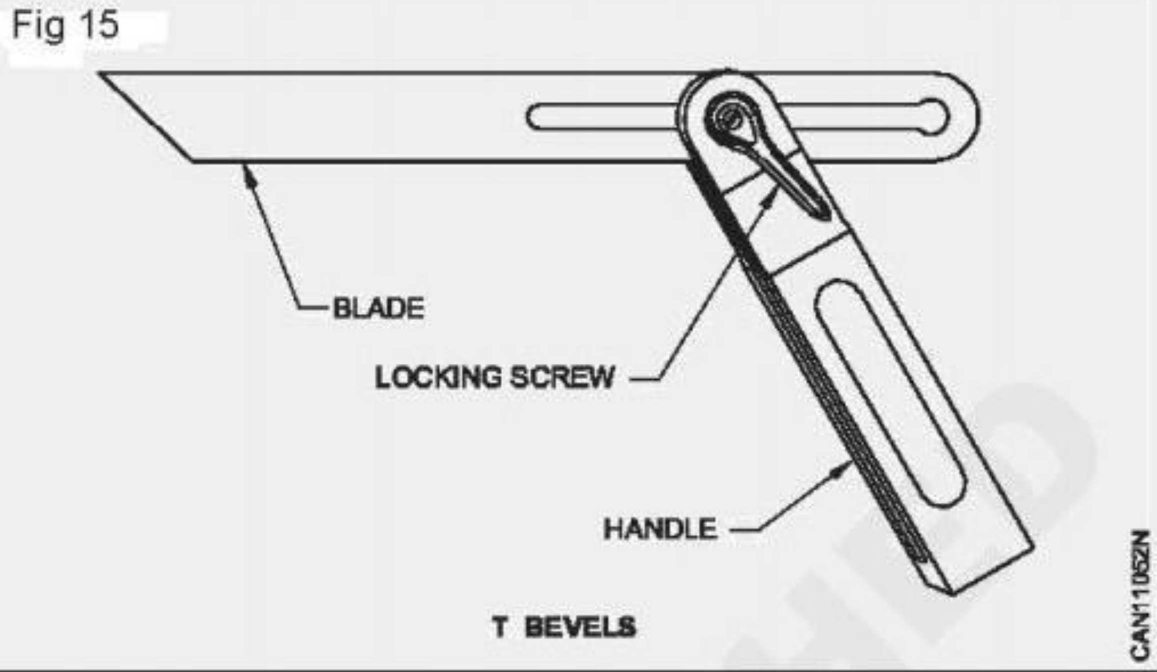
This is made of hard wood and has an adjusting screw in the end of the beam.

.

'T' bevel or bevel square

The T- bevel is used to test and transfer angles other than right angles. The bevel is called sliding bevel because it has an adjustable sliding blade. The blade may be locked by a wing nut or set screw.

Fig 15



T BEVELS

Uses

The sliding bevel is used for laying out dovetails, side rails for chairs, chamfers, bevels and for transferring angles from the drawing to the work piece. The Parts of the bevel square. (Fig 15)

Handle: Handle is made of hard wood, cast iron and aluminum. The top edge is half rounded and there is a slot to fix the bevel square

Blade: One edge of the blade is half rounded and their other edge is cut at 45°. There is a longitudinal slot. The handle is fitted with the wing nut in the slot or with a machine screw. The slotted blade passes through a slot in the stock.

Uses

The sliding bevel is used for laying out dovetails, side rails for chairs, chamfers, bevels and for transferring angles from the drawing to the work piece. The Parts of the bevel square. (Fig 15)**Locking nut:** This may be a wing nut or a set screw used for loosening or tightening the blade

Type of bench vice and their uses

Objectives: At the end of this lesson you shall be able to
- state the constructional features various types of bench vices
- explain the uses quick release vice and saw vice
- brief the uses of bench vice.

Wood must be held steadily. If it is to be accurately sawn, chiselled and planed

- For this reason carpenter swork bench consist different types of vice
- Most commonly fitted bench vice consist of two metal Jaws to hold the work
- One Jaw of the vice is fixed to the work bench
- The other Jaw is movable parallel to the fixed jaw
- To operate this there is a threaded shaft and a handle.
- Two wooden blocks are used inside the Jaws to protect the work from damage

There are three types of vice

1 Quick release vice (or) wood worker s vice.

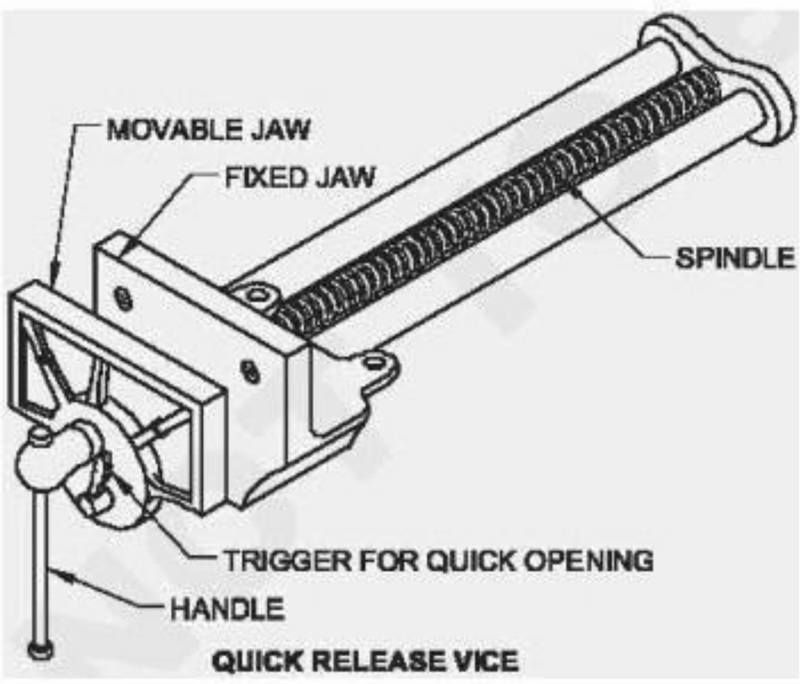
2 Saw vice

2 Bench vice

Quick release vice

- In the quick release vice moveable jaw is quickly released and get clamped with fixed Jaw
- A box nut is provided in its threaded shafts for the quick release system (Fig 1)

Fig 1



The jaw is made of cast iron

- The threaded shaft is made of steel
- The vice is specified by the width of the Jaw Precautions

1 Vices should not be used as ANVIL

2 The thread shaft and box nut should lubricated.

Introduction of different saws and their uses

Objectives: At the end of this lesson you shall be able to
- **state the various types of** saws
- **brief the specific uses of straight cutting saws.**

There are several types of saws for cutting wood. some are used to make straight cuts and others are used make curved cuts. As its name suggests, a cross cut saw is used cut wood perpendicular or at an angle to the wood grain.

Types of Saw

1 Straight cutting saws

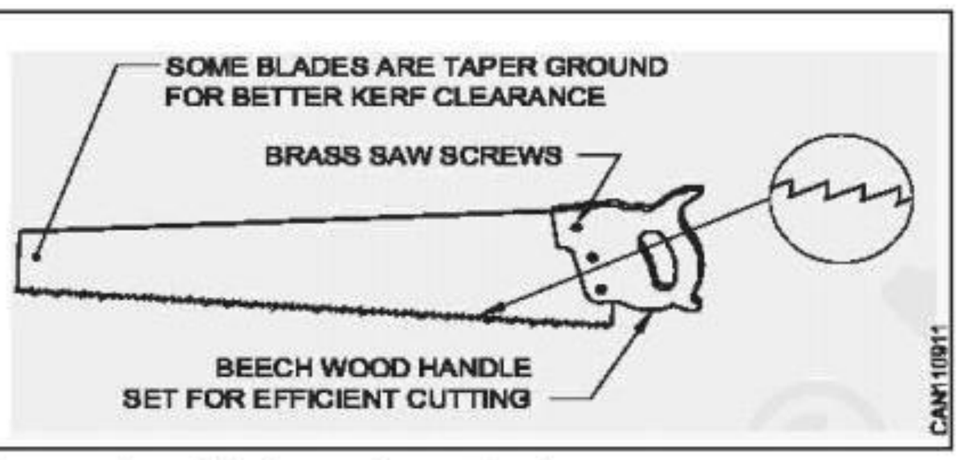
2 Curve cutting saws (or) special saws.

Straight cutting saws

1 Rip saw	2 Cross cut saw
3 Hand saw	4 Panel saw
5 Tenon saw	6 Dove tail saw

Rip Saw (Fig 1)

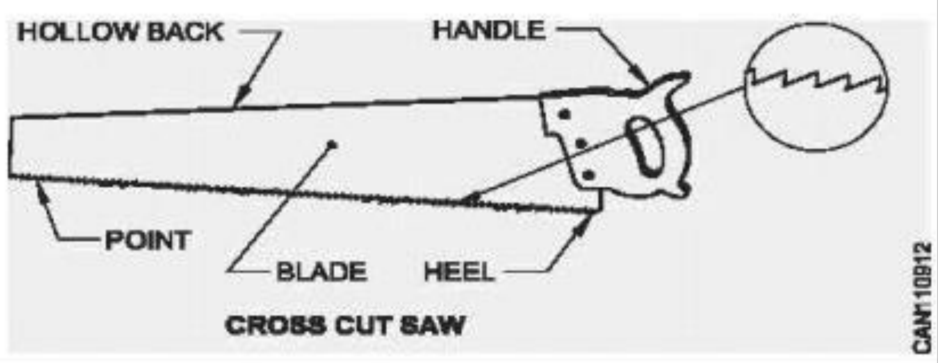
Fig 1



— Blade is made of thin spring steel
- Used for sawing along the grain
- Blade is fixed to the wooden handle by riveting or screwing
- Teeth of rip saw vary in size as per the need of the fitness of work to be done
- Handle is made of beech (or) apple wood
- It has two teeth per centimetre length
- length of the blade is 60 to 70cm
- Specified by its length
- Teeth angle is less than 90^0
- It has 3 to 6 teeth per 25mm.

Cross cut saw(Fig 2)

Fig Z



- Used for cutting across the grains of timber
- Length of blade is 56 to 70cm
- It has 5 to 9 teeth per 25mm 28

Construction Related Theory for Exercise 1.1.15
Carpenter - Safety Precautions, Hand Tools and Timber

Wood working planes

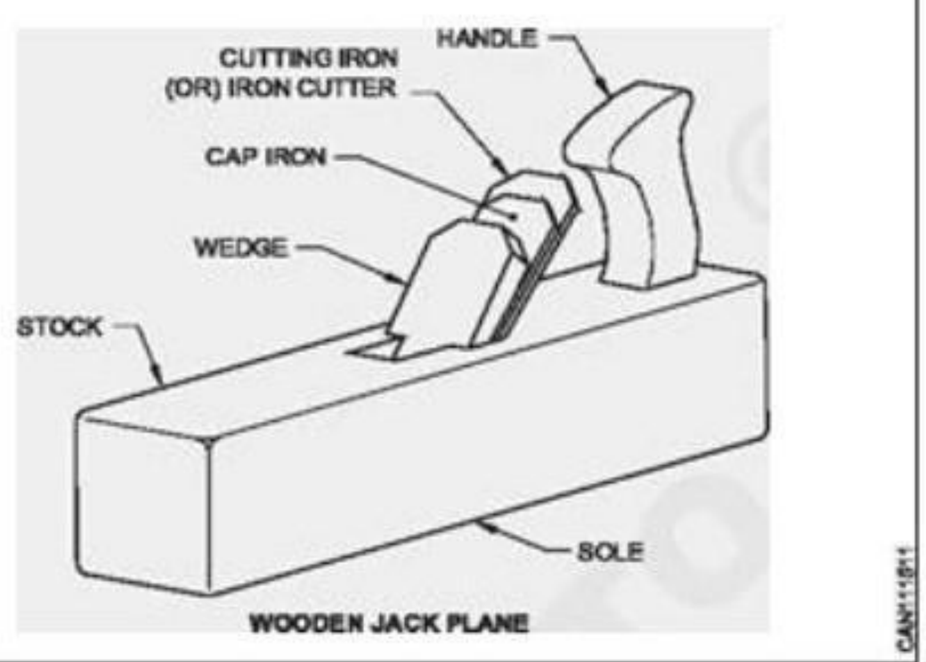
Objectives : At the end of this lesson you shall be able to
- name the various types of planes
- state the uses of the various planes
- state **constructional** feature of various planes
- **specify the** planes.

A plane is a hand tool used for smoothing or shaping pieces of wood.

Common types are

1 Jackplane
2 Smoothing plane
3 Trying plane
4 Toothing plane
5 Finishing plane
6 Router plane
7 Plough plane
8 Adjustable metal jack plane.

Fig1



1 This plane is used for planing the job to size quickly and truly. Stock is made of wood or steel. Handle is fixed behind the cutting iron

The size is 375 mm x 66mm x 47mm. The angle of cutting iron is 45° and the cutting iron is sharpened in a curve. The mouth of the plane is big enough to accommodate thicker wood shavings. The cutting iron further projects outside than in other planes.

2 **Smoothing** plane (Fig 2): This is used when the surface has to be planed further to smoothness.

- The size is 240 x 66 x 65mm
- The width of cutting iron is 48mm
- The cutting edge is sharpened slightly oval across the iron. The angle of the cutting iron is 30°. The wooden smoothing plane has no handle.

1 Jackplane (Fig1)